TEST CAMPAIGNS · DP STEEL SHEET

Specimens per geometry and orientation to the rolling direction, one material. Beerli et al. 2026; shear: 3 geometries (Roth & Mohr 2018), 2 repeats.

not sure · Q = question below

SLOW RATE · 84 specimens

UT 0.001 /s · others 6.6e-6 m/s

GEOMETRY	0°	15°	30°	45°	60°	75°	90°	TOTAL	Q
TENSION-TYPE · waterjet									
UT40	3	3	3	3	3	3	3	**21**	
NT20	3			3			3	**9**	
NT06	3			3			3	**9**	
CHD48	3			3			3	**9**	Q1 Q6
MNAK_RECT_FULL	3			3			3	**9**	Q5
MNAK_RECT_HOLE	3			3			3	**9**	Q5
SHEAR · wire EDM, 2 repeats									
SHLD	2			2			2	**6**	Q4
SHMD	2			2			2	**6**	Q4
SHHD	2			2			2	**6**	Q4
TOTAL	**24**	**3**	**3**	**24**	**3**	**3**	**24**	**84**	

MEDIUM RATE · 18 specimens

UT 1 /s · others 6.6e-3 m/s

GEOMETRY	0°	TOTAL	Q
TENSION-TYPE · waterjet			
UT40	3	**3**	Q2
NT20	3	**3**	Q2
NT06	3	**3**	Q2
CHD48	3	**3**	Q1 Q2
SHEAR · wire EDM, 2 repeats			
SHLD	2	**2**	Q4
SHMD	2	**2**	Q4
SHHD	2	**2**	Q4
TOTAL	**18**	**18**	

HIGH RATE · 18 specimens

UT 100 /s · others about 1.4 m/s

GEOMETRY	0°	TOTAL	Q
TENSION-TYPE · waterjet			
UT19	3	**3**	Q2
NT20	3	**3**	Q2
NT06	3	**3**	Q2
CHD48	3	**3**	Q1 Q2
SHEAR · wire EDM, 2 repeats			
SHLD	2	**2**	Q4
SHMD	2	**2**	Q4
SHHD	2	**2**	Q4
TOTAL	**18**	**18**	

ALL RATES: 84 + 18 + 18 = 120 SPECIMENS

QUESTIONS

Q1 Which steel grade? (CH5 up to DP1000, CH8 for DP1180 / DP1470.)

Q2 Medium and high rate: 3 repeats per tension-type specimen?

Q3 Medium and high rate: 0° only?

Q4 Shear: also 135°? The 3 geometries (LD / MD / HD) also at medium and high rate?

Q5 MNAK_RECT_FULL = punch and MNAK_RECT_HOLE = mini-Nakazima? 0, 45, 90° × 3 each?

Q6 Central hole: is the 4.8 mm hole reamed to 5 mm after cutting? Is CH8 needed?